Lab 5 — Steady State and Eigen

Friday, November 20 (~20 min)

Last updated: October 29, 2026.

In class Friday, November 20 (same day as Lecture 37 / Quiz 10). Quiz 10 is Week 11 inverses / rank / determinants — **not** this lab.

Credit: Lab points are for effort in class, not for a Canvas file upload. Preview the one-pager; write on the hard copy handed out in class.

Payoff. Lab 4 asked why different starts share a long-run mix. Today: that mix is the $\lambda = 1$ **eigenvector** (scaled); the other eigenvalue has $|\lambda| < 1$, so its mode fades. Same story as Lecture 37's dominant-eigenbehavior fill-in.

Learning objectives

1. Find a steady state of $x = Ax$ (or an eigenvector for $\lambda = 1$).
2. Use software eigenvalues/vectors and match them to the Lab 4 long-run mix.
3. State how an eigen-direction explains repeated multiplication by A ($A^t v = \lambda^t v$).

Math connections

Eigenvalues and eigenvectors (Lectures 35–37); build on Lab 4's iteration.

Key syntax (this lab)

Task	Octave	Python
Eigenpairs	`[V,D]=eig(A)`	`lam, V = np.linalg.eig(A)`
Diagonal entries	`diag(D)`	`lam`
Normalize a vector	`v/sum(v)` or `v/norm(v)`	same idea with NumPy

Ops strip

Same campus matrix as Lab 4:

$$A = \begin{bmatrix} 0.7 & 0.2 \\ 0.3 & 0.8 \end{bmatrix}.$$

C5.0. Solve $(A - I)x = 0$ (or ask software for a nullspace / eigenvector for $\lambda = 1$). Scale so the two entries sum to 1000 (total students).

Expected: a multiple of $(2, 3)$ — e.g. $x = (400, 600)$ when entries sum to 1000.

Application

C5.1. Compute all eigenvalues of A in software. Which one has magnitude 1? Which has $|\lambda| < 1$?

Expected: $\lambda_1 = 1$ and $\lambda_2 = 0.5$ (order may differ).

C5.2. Take the eigenvector for $\lambda = 1$, scale so entries sum to 1000, and compare to your Lab 4 x_{20}. One sentence: why does iteration settle there?

Expected: the long-run state is (essentially) that $\lambda = 1$ direction; other modes shrink because $|\lambda_2| < 1$.

C5.3. Capstone preview: when you compress a matrix with SVD, you keep the **largest** singular values. In one sentence, how is "keep the big directions" similar to "the $\lambda = 1$ mode wins the long run"?

Expected: both stories pick the directions that dominate the object (image residual vs. iterated state).

One-pager

Preview the Lab 5 ticket; write on the hard copy in class.